PVSYST
PHOTOVOLTAIC
SOFTWARE

PVsyst TRIAL

PVsyst - Simulation report

Grid-Connected System

Project: Guaira

Variant: New simulation variant
No 3D scene defined, no shadings
System power: 3000 Wp
Guaira - Brazil

Author

PVsyst V8.1.7
VC0, Simulation date:
16/10/26 21:49
with V8.1.7

Project summary

Geographical Site
Guaira
Brazil

Situation

Latitude	-24.0852 °(S)
Longitude	-54.2568 °(W)
Altitude	270 m
Time zone	UTC-3

Project settings
Albedo 0.20

Weather data
Guaira
NASA POWER satellite data 2001-2020 - Synthetic

System summary

Grid-Connected System

No 3D scene defined, no shadings

Orientation #1
Horizontal plane

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

System information

PV Array		Inverters	
Nb. of modules	5 units	Nb. of units	1 unit
Pnom total	3000 Wp	Total power	2.5 kWac
		Pnom ratio	1.20

Results summary

Simulation step	Hourly				
Produced Energy	4574.1 kWh/year	Specific production	1525 kWh/kWp/year	Perf. Ratio PR	84.68 %

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General parameters

Grid-Connected System — **No 3D scene defined, no shadings**

Simulation step
Hourly

Orientation #1
Horizontal plane

Models used

Transposition	Perez
Diffuse	Perez, Meteonorm
Circumsolar	separate

Horizon
Free Horizon

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer	Generic
Model	JKM-600N-72HL4-V
(Original PVsyst database)	
Unit Nom. Power	600 Wp
Number of PV modules	5 units
Nominal (STC)	3000 Wp
Modules	1 unit x 5 In series
At operating cond. (50°C)	
Pmpp	2783 Wp
U mpp	204 V
I mpp	14 A

Inverter

Manufacturer	Generic
Model	MIC 2500TL-X
(Original PVsyst database)	
Unit Nom. Power	2.50 kWac
Number of inverters	1 unit
Total power	2.5 kWac
Operating voltage	65-550 V
Pnom ratio (DC:AC)	1.20

Total PV power

Nominal (STC)	3.00 kWp
Total	5 modules
Module area	12.9 m²
Cell area	11.9 m²

Total inverter power

Total power	2.5 kWac
Number of inverters	1 unit
Pnom ratio	1.20

Array losses

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 245 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss
Loss Fraction -0.75 %

Module mismatch losses
Loss Fraction 2.00 % at MPP

IAM loss factor
Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000

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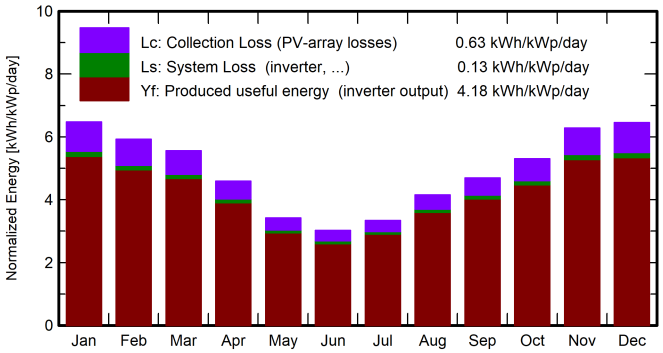
Main results

Simulation step
Hourly

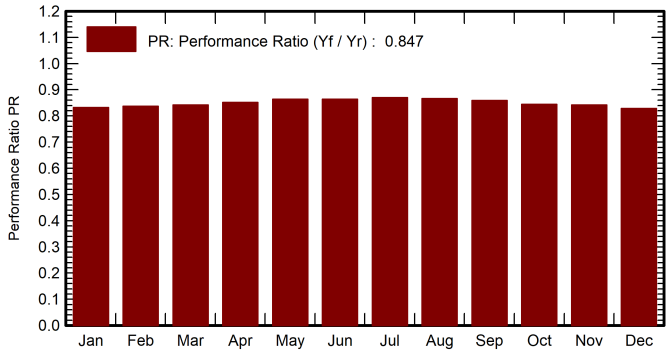
System Production

Produced Energy	4574.1 kWh/year	Specific production	1525 kWh/kWp/year
		Perf. Ratio PR	84.68 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	200.8	76.04	27.06	200.8	196.2	516.3	501.4	0.832
February	166.0	65.63	26.84	165.9	162.0	428.8	416.4	0.837
March	172.4	59.14	26.23	172.3	167.7	448.2	435.4	0.842
April	138.0	44.86	23.85	137.9	133.5	362.7	352.2	0.851
May	106.2	39.14	19.41	106.2	101.8	283.6	275.1	0.864
June	90.7	34.76	17.93	90.7	86.6	242.7	235.1	0.864
July	103.7	35.43	17.40	103.7	99.4	279.0	270.6	0.870
August	129.0	40.43	19.14	128.9	124.4	344.9	335.0	0.866
September	141.0	47.66	21.85	140.9	137.1	374.3	363.3	0.859
October	164.5	62.19	24.70	164.4	160.4	428.9	416.3	0.844
November	188.7	63.79	25.41	188.6	184.3	490.0	476.1	0.841
December	200.3	76.47	26.65	200.1	195.4	511.9	497.4	0.828
Year	1801.4	645.53	23.02	1800.5	1748.8	4711.2	4574.1	0.847

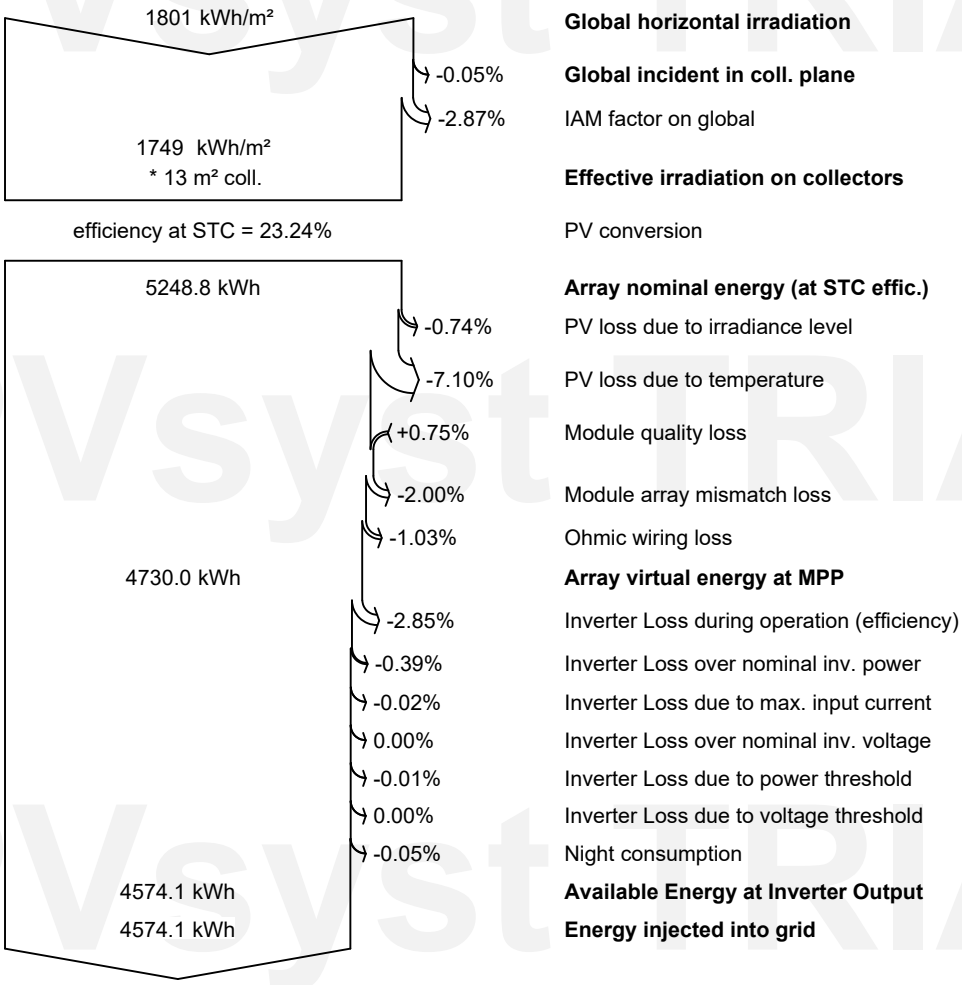
Legends

GlobHor	Global horizontal irradiation
DiffHor	Horizontal diffuse irradiation
T_Amb	Ambient Temperature
GlobInc	Global incident in coll. plane
GlobEff	Effective Global, corr. for IAM and shadings
EArray	Effective energy at the output of the array
E_Grid	Energy injected into grid
PR	Performance Ratio

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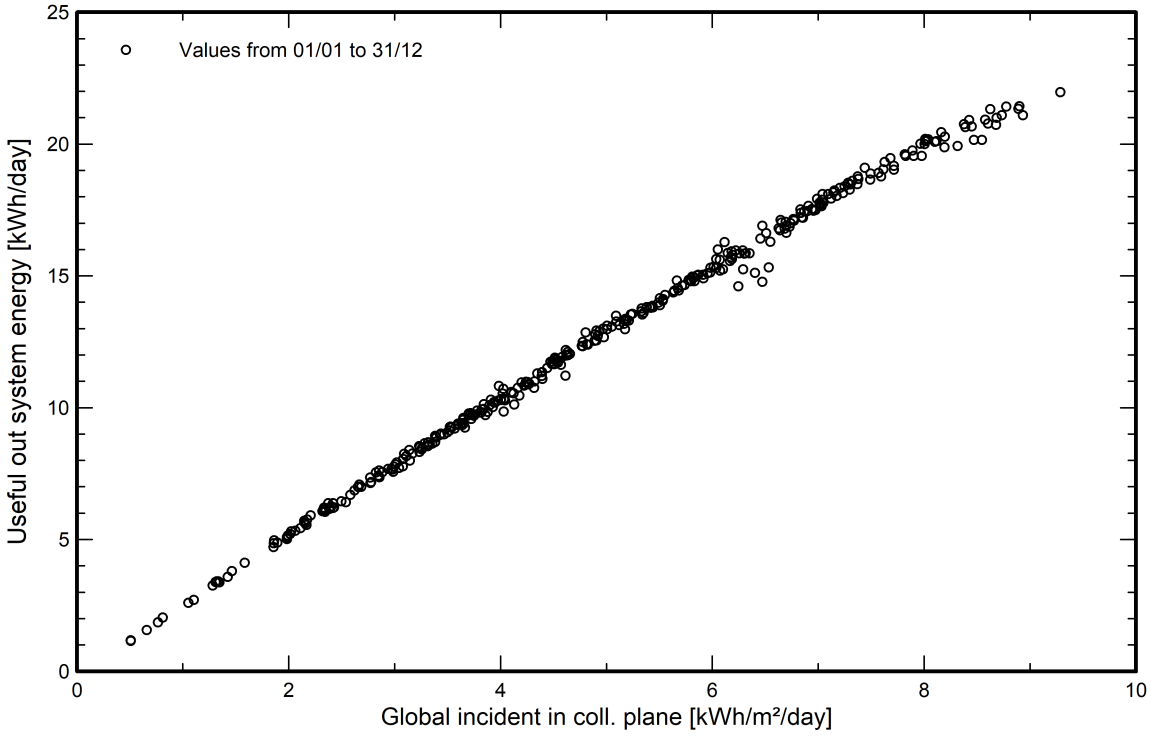
Loss diagram



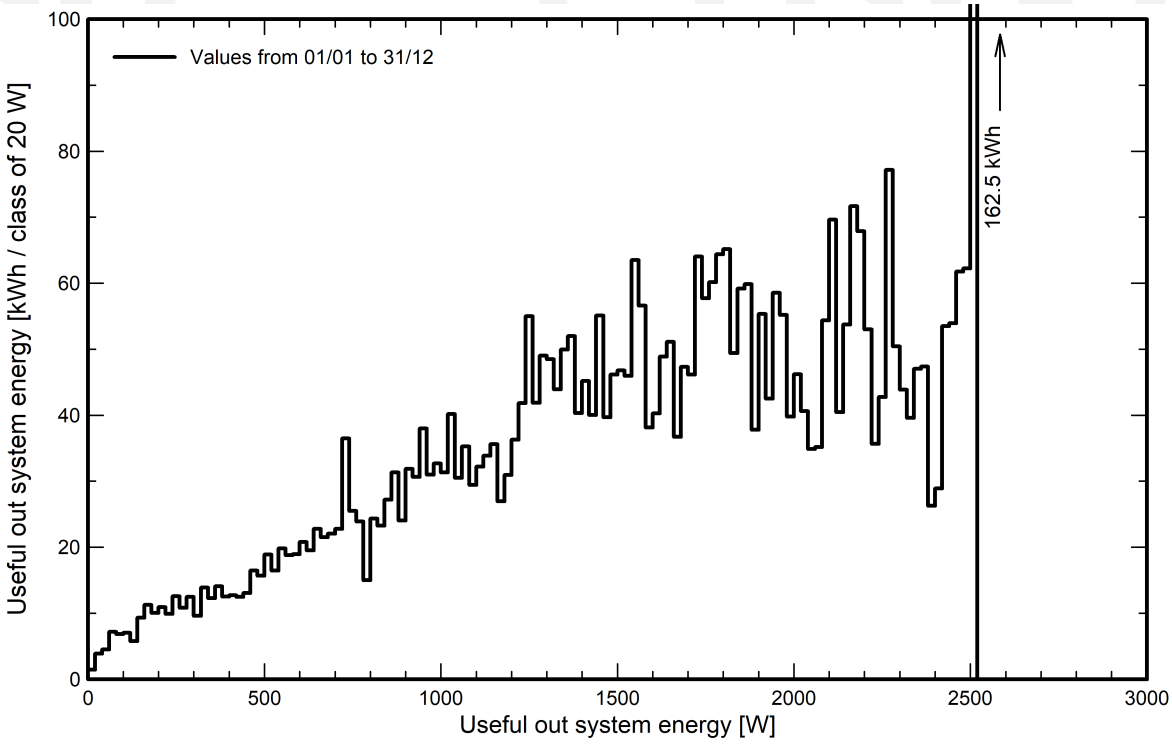
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Predef. graphs

Daily Input/Output diagram

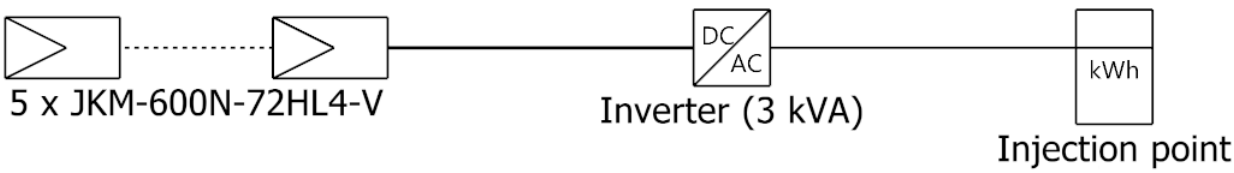


System Output Power Distribution



Single-line diagram

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PV module	JKM-600N-72HL4-V
Inverter	MIC 2500TL-X
String	5 x JKM-600N-72HL4-V

	Guaira	
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